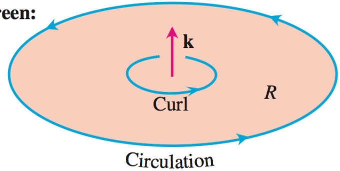
Topic: Stokes' Theorem

I. Stokes' Theorem

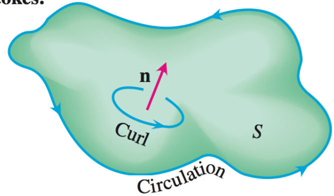
Thm: Let S be a surface with boundary C and $\mathbf{F}(x, y, z)$ be a vector-valued function defined on S. Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dA = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{n}\, dA$$

Green:



Stokes:



Note that Green's Theorem is a special case of Stokes' theorem when S is a plane region R. In this case, $\mathbf{n}$ is just $\mathbf{k} = [0, 0, 1]$ and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dA = \iint_R (\operatorname{curl} \mathbf{F}) \cdot \mathbf{k}\, dA = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Ex: $\mathbf{F} = [-5y, 4x, z]$, and C is the circle $x^2 + y^2 = 16$, $z = 4$. Find $\displaystyle\oint_C \mathbf{F} \cdot d\mathbf{r}$

Note: curl $\mathbf{F} = [0, 0, 9]$ and $\mathbf{n} = [0, 0, 1]$

So by Stokes' Theorem and the fact S is a circle

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dA = \iint_S 9\, dA = 9 \cdot Area(S) = 9(\pi \cdot 4^2) = 144\pi$$

Let $\mathbf{F} = [4yz\,,\, 6z - 3xy\,,\, x]$. Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for each of the following curves:

a. C is the union of the straight paths from $(0,0,0)$ to $(0,0,5)$ to $(0,3,5)$ and back to $(0,0,0)$

b. C is the union of the straight paths from $(0,0,5)$ to $(2,0,0)$ to $(0,5,0)$ and back to $(0,0,5)$

$$\text{curl}\,\mathbf{F} = [-6,\, 4y-1\,,\, -3y-4z] \quad \& \quad \mathbf{n} = [-1,0,0]$$

$$\Longrightarrow \qquad \text{curl}\,\mathbf{F} \cdot \mathbf{n} = 6$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl}\,\mathbf{F}) \cdot \mathbf{n}\, dA$$

$$= \iint_S 6\, dA$$

$$= 6 \cdot Area(S)$$

$$= 6\left[\frac{1}{2}(5)(3)\right] = 45$$

curl $\mathbf{F} = [-6, 4y - 1, -3y - 4z]$ & $\mathbf{n} = ?$

Need the Eqn of the plane which passes through: $(2,0,0), (0,5,0), (0,0,5)$

General Form: $ax + by + cz = d$

By observation of where the plane intersects with each axis, we can determine that the equation of the plane must be $5x + 2y + 2z = 10$

Setting $u = x$ and $v = y$ we have:

$$\mathbf{r}(u,v) = [u, v, 5 - \frac{5}{2}u - v]$$

$\mathbf{N} = [5, 2, 2]$ (coefficients of the plane eqn)

$$\begin{aligned} \implies \quad \text{curl}(\mathbf{F}) \cdot \mathbf{N} &= -30 + 8y - 2 - 6y - 8z \\ &= -32 + 2y - 8z \\ &= -32 + 2v - 8\left(5 - \tfrac{5}{2}u - v\right) \\ &= -72 + 20u + 10v \end{aligned}$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl}\,\mathbf{F}) \cdot \mathbf{n}\, dA = \iint_R \text{curl}\,\mathbf{F} \cdot \mathbf{N}\, du dv \quad = \int_0^2 \int_0^{-\frac{5}{2}u+5} -72 + 20u + 10v\, du dv$$

Ex: $\mathbf{F} = [-y, 2z, 0]$, C is the boundary of side of the cylinder $y^2 + z^2 = 4$, with $z \geq 0$, $0 \leq x \leq 3$ and oriented outward. Graph the surface and use Stokes' thm to compute $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$.

curl $\mathbf{F} = [-2, 0, 1]$ & $\mathbf{n} = ?$

Using polar coordinates

$$x = x$$

$$y = 2\cos\theta \quad \underset{v=\theta}{\overset{u=x}{\Longrightarrow}} \quad \mathbf{r}(u, v) = [u, 2\cos v, 2\sin v]$$

$$z = 2\sin\theta \qquad 0 \leq u \leq 3,\ 0 \leq v \leq \pi$$

$$\mathbf{r}_u = [1, 0, 0]$$

$$\mathbf{r}_v = [0, -2\sin v, 2\cos v]$$

$$\mathbf{N} = [0, -2\cos v, -2\sin v]$$

Note: With this parameterization, our orientation is reversed.

$$\Longrightarrow \quad \mathbf{N} = [0, 2\cos v, 2\sin v]$$

$$\Longrightarrow \quad \text{curl}\,\mathbf{F} \cdot \mathbf{N} = 2\sin v$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl}\,\mathbf{F}) \cdot \mathbf{n}\, dA = \iint_R \text{curl}\,\mathbf{F} \cdot \mathbf{N}\, du dv = \int_0^\pi \int_0^3 2\sin v\, du dv = 12$$